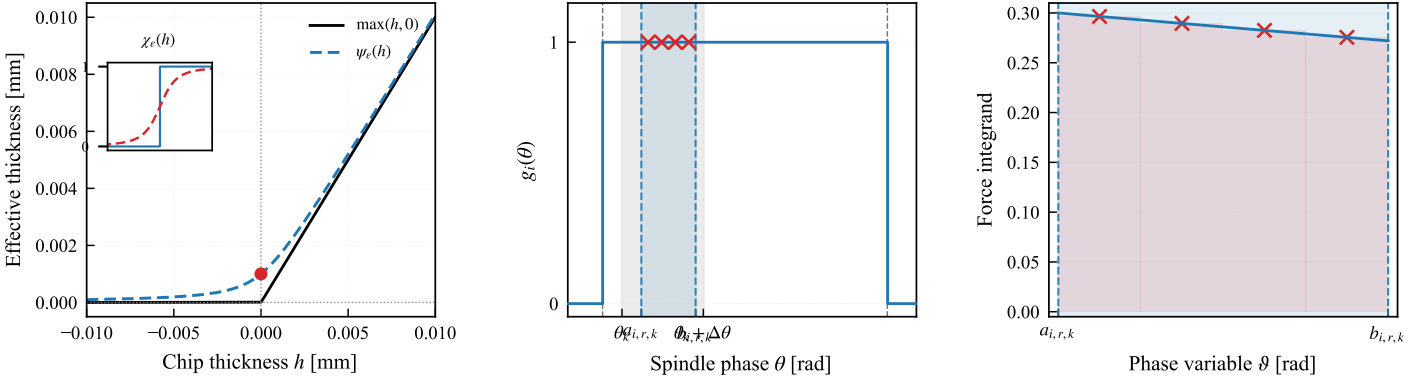
(a) Chip-loss
Effective thickness [mm]
Chip thickness h [mm]
$\max(h, 0)$
$\psi_\varepsilon(h)$
$\chi_\varepsilon(h)$
(b) Phase-cell engagement
$g_i(\theta)$
Spindle phase θ [rad]
(c) Engagement quadrature
Force integrand
Phase variable ϑ [rad]